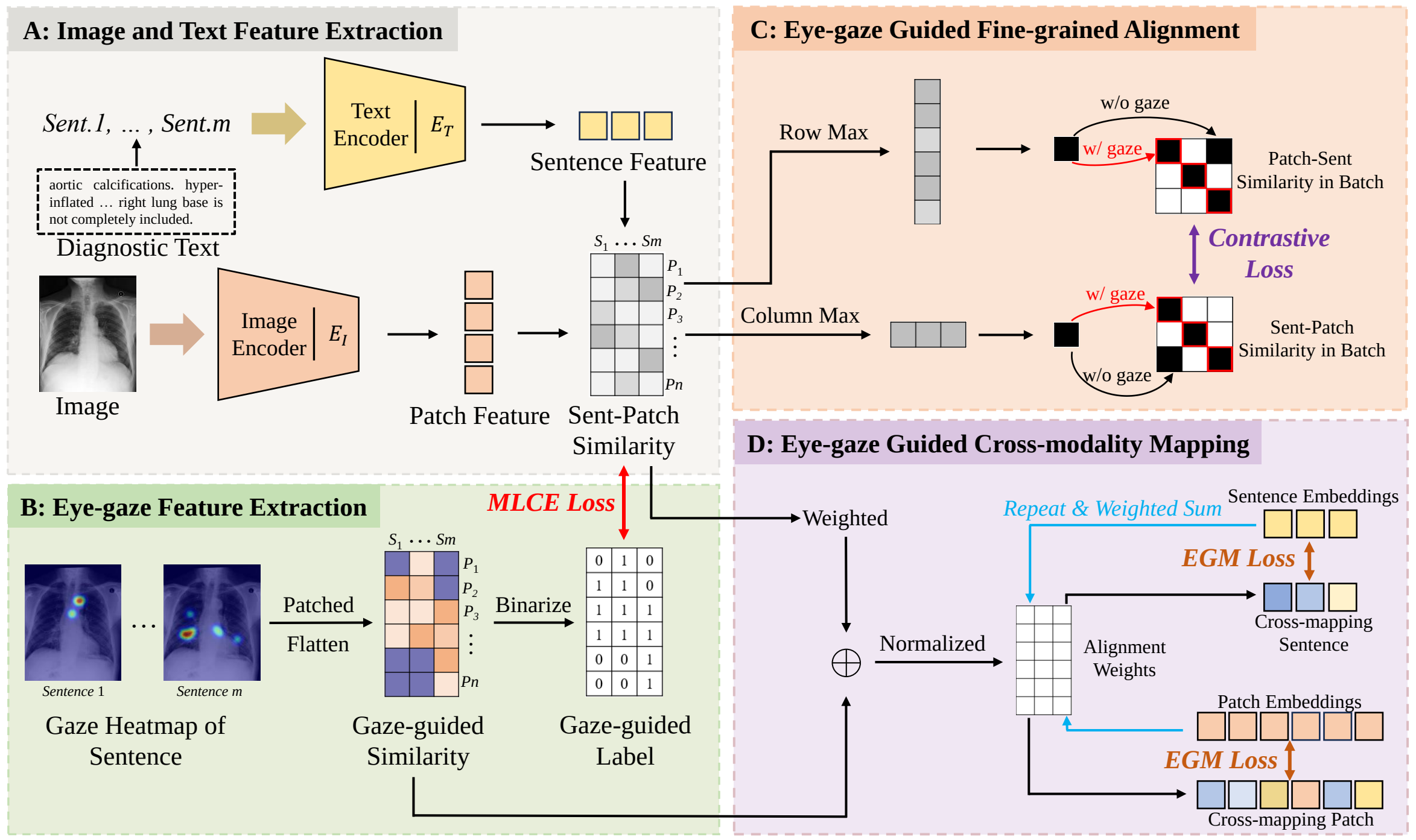

A: Image and Text Feature Extraction
Sent.1, … , Sent.m
aortic calcifications. hyper-inflated … right lung base is not completely included.
Diagnostic Text
Text Encoder
E_T
Sentence Feature
Image
Image Encoder
E_I
Patch Feature
S_1 … Sm
P_1
P_2
P_3
Pn
Sent-Patch Similarity
C: Eye-gaze Guided Fine-grained Alignment
Row Max
Column Max
w/o gaze
w/ gaze
Patch-Sent Similarity in Batch
Contrastive Loss
w/ gaze
w/o gaze
Sent-Patch Similarity in Batch
B: Eye-gaze Feature Extraction
MLCE Loss
Sentence 1
Sentence m
Gaze Heatmap of Sentence
Patched
Flatten
Binarize
Gaze-guided Similarity
Gaze-guided Label
D: Eye-gaze Guided Cross-modality Mapping
Weighted
Normalized
Repeat & Weighted Sum
Sentence Embeddings
EGM Loss
Cross-mapping Sentence
Alignment Weights
Patch Embeddings
EGM Loss
Cross-mapping Patch